BUSINESS INTELLIGENCE LAB

UNIT 3 | SESSION 10

EXPERIMENT 10

BI Deployment, Monitoring and Backup & Restore using Microsoft Power BI

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1. AIM

To understand and design Business Intelligence deployment, monitoring, backup, and recovery strategies using Microsoft Power BI for ensuring availability, reliability, and continuity of BI solutions.

2. OBJECTIVES

- To identify the BI assets created in Experiments 5 to 9 and plan their deployment.
- To design a DEV and PROD workspace structure with clear roles and governance rules.
- To deploy the Global Superstore BI solution to end users through a Power BI app.
- To monitor report performance, dataset refresh status, usage and lineage.
- To prepare a Monitoring Matrix with metrics, frequency and responsible personnel.
- To design a backup schedule and restore procedure for BI assets.
- To analyze and test a disaster recovery scenario for an accidentally deleted executive dashboard.

3. TOOLS / SOFTWARE REQUIRED

1. Microsoft Power BI Desktop (Publish, Performance analyzer, Save As, Export as template).
2. Microsoft Power BI Service (workspaces, Manage access, apps, refresh history, usage metrics, lineage view, recycle bin).
3. Global Superstore Dataset (Global Superstore 2018.xlsx).
4. Microsoft Excel for the deployment, monitoring and backup matrices.
5. Microsoft OneDrive for storing backup files.

4. BI ENVIRONMENT USED

The experiment uses the same Global Superstore data model and report developed in Experiments 5 to 9, saved and published as **BI Expe-10**. The data source is **Global Superstore 2018.xlsx**, connected to the Power BI Service through a gateway.

Environment item	Name
Report and semantic model	BI Expe-10 (8 report pages)
Development workspace	Global Superstore BI - DEV
Production workspace	Global Superstore BI

Environment item	Name
Power BI app	Global Superstore Analytics
Backup location	OneDrive: Business_Intelligence_Tools / Experiment 10 / BI_Backups

5. THEORY

5.1 BI Deployment

BI Deployment refers to making reports, dashboards, and analytical solutions available to end users. In Power BI, content is developed in Power BI Desktop, published to a development workspace for testing, promoted to a production workspace, and distributed to business users through apps, sharing, or subscriptions. Separating development and production prevents untested changes from reaching decision makers.

5.2 Monitoring

Monitoring involves tracking system performance, usage, refresh status, and operational health. Power BI provides several monitoring tools:

- **Performance analyzer:** measures how long each visual takes to load in Power BI Desktop.
- **Refresh history:** shows the status, time and error message of every dataset refresh.
- **Usage metrics report:** shows views, unique viewers and most-viewed pages over the last 90 days.
- **Lineage view:** shows dependencies between data sources, semantic models, reports and apps.

5.3 Backup and Restore

Backup and Restore mechanisms protect BI assets from accidental deletion, corruption, or system failures. Power BI backup options include versioned .pbix files (report, model and data), .pbit templates (model, measures, roles and pages without data), downloading the .pbix from the Service, storing backups in OneDrive or SharePoint with version history, and the workspace Recycle bin for restoring deleted items.

5.4 Business Continuity Measures

Term	Meaning	Target in this design
RPO (Recovery Point Objective)	Maximum acceptable data or work loss, measured in time	1 day (daily backup)
RTO (Recovery Time Objective)	Maximum acceptable time to restore the service	1 hour for executive content
Availability	Share of time the BI solution is usable by business users	Executive dashboards available every working day

5.5 Business Scenario

Global Superstore has successfully developed reports, dashboards, analytical solutions, security policies, and automated reporting mechanisms. The management team now wants to deploy the BI solution across the organization, monitor its usage and performance, and establish backup and recovery procedures. Acting as BI Solution Architects, a complete deployment and maintenance strategy is designed and demonstrated in Power BI.

6. PROCEDURE

Part A: BI Deployment Planning

Step 1: Identified BI assets created in previous experiments.

Experiment	BI asset	Type
Experiment 5	Geographical Sales Analysis Dashboard, Relational Report, Matrix Report, Comparative Analysis, Interactive Dashboard	Report pages
Experiment 6	Executive Dashboard, Business Scorecard (targets and achievement)	Dashboard and scorecard pages
Experiment 7	Business Insights page, Business Insight Report, Executive Presentation	Analytics page and documents
Experiment 8	Published report, email subscriptions, data alerts, Sales Change % and Returns Change % measures	Automated reporting
Experiment 9	Global Superstore BI workspace, workspace roles, 5 Row-Level Security roles	Security configuration
All	Global Superstore semantic model (Fact_Sales, Dim tables, DAX measures)	Dataset

Step 2: Listed the items that require deployment.

Item	Name	Deployed to
Report	BI Expe-10 (8 pages)	DEV, then PROD
Semantic model	BI Expe-10 with DAX measures and RLS roles	DEV, then PROD
Dashboards	Executive Dashboard, Business Scorecard, Business Insights	PROD, through the app
Workspaces	Global Superstore BI - DEV, Global Superstore BI	Power BI Service
App	Global Superstore Analytics	Management audience
Data source	Global Superstore 2018.xlsx	Connected through gateway

Step 3: Created the BI Deployment Plan.

Phase	Activity	Output	Owner
1. Prepare	Finalize the report in Desktop, check measures, RLS roles and page layout	BI_Experiment_10.pbix	BI Developer
2. Deploy to DEV	Publish to Global Superstore BI - DEV	DEV report and model	BI Developer
3. Test	Check all 8 pages, slicers, KPIs and RLS (View as / Test as role)	Test sign-off	Tester / QA
4. Deploy to PROD	Publish the approved file to Global Superstore BI	PROD report and model	BI Administrator

Phase	Activity	Output	Owner
5. Distribute	Create and publish the Global Superstore Analytics app with an audience	Live app link	BI Administrator
6. Operate	Monitor performance, refresh and usage; run backups	Monitoring and backup records	BI Administrator

Step 4: Identified intended users and departments.

Department	Users	Content needed	Access method
Executive	CEO	Executive Dashboard, Business Scorecard, Business Insights	App
Finance	Finance Manager	Scorecard, profit and margin analysis	App
Sales	Sales Manager	Sales trends, customers, regional sales	App
Regional Management	5 Regional Managers	Regional pages filtered by RLS	App (Viewer + RLS)
IT / BI	BI Developer, Tester, BI Administrator	All content for development and support	DEV and PROD workspaces

Step 5: Defined deployment objectives and expected outcomes.

Objective	Expected outcome
Single trusted source of reports	All departments use the same approved PROD content
Controlled releases	No untested change reaches management; every change passes DEV testing
Easy access for users	Users open one app link instead of searching workspaces
Secure distribution	Workspace roles and RLS limit each user to the right content and data
Reliable operations	Refresh failures and slow visuals are detected early through monitoring
Business continuity	Deleted or damaged content can be restored within the 1 hour RTO

Part B: Workspace and Access Planning

Step 6: Designed the Power BI workspace structure. Two workspaces were created: a development workspace for building and testing, and a production workspace for approved content. Business users receive content only through the app published from production.

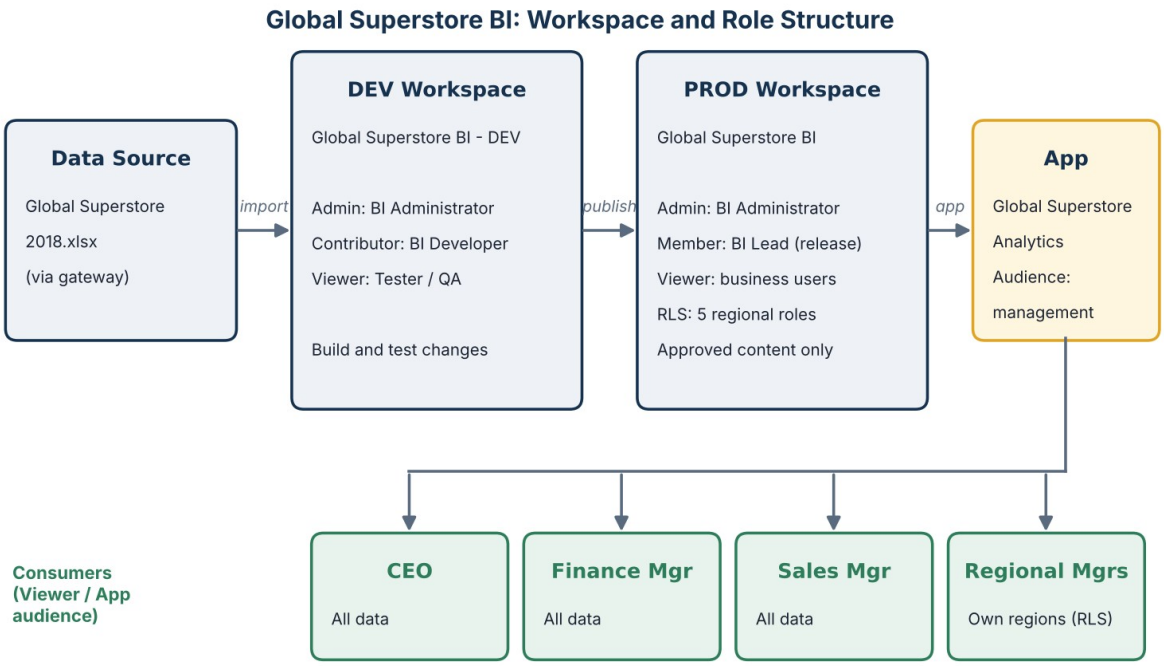


Figure 1: Workspace and Role Structure Diagram for the Global Superstore BI solution.

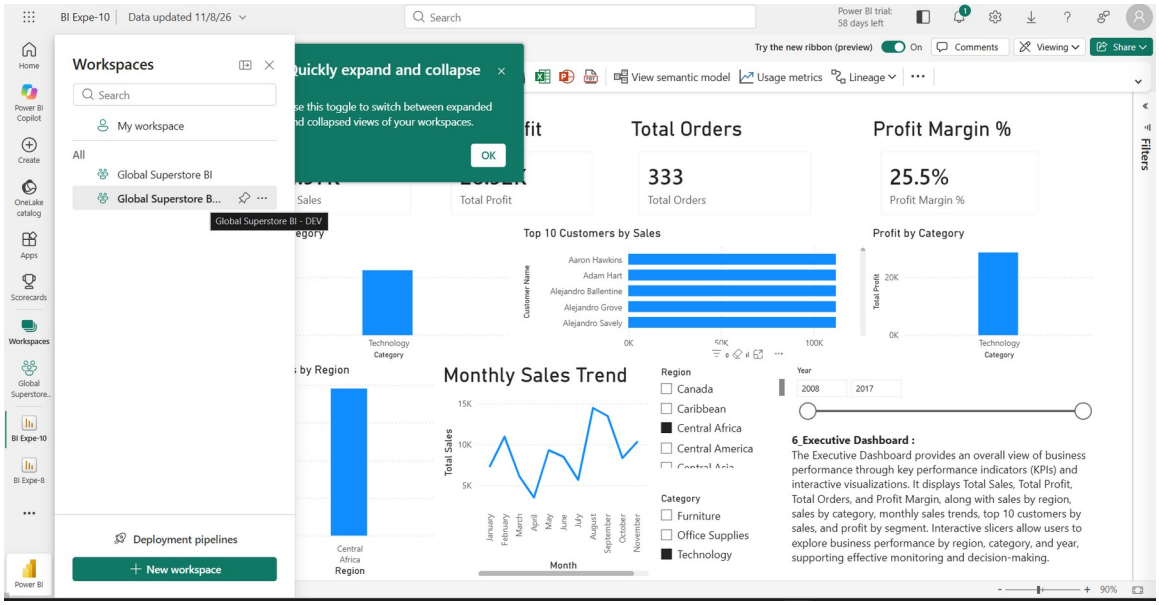


Figure 2: Power BI Service: Global Superstore BI (PROD) and Global Superstore BI - DEV workspaces, with BI Expe-10 published.

The report was first published to DEV and checked page by page, then published to PROD. The app was then created from PROD (Create app), with the report added as content and access granted to a specific audience.

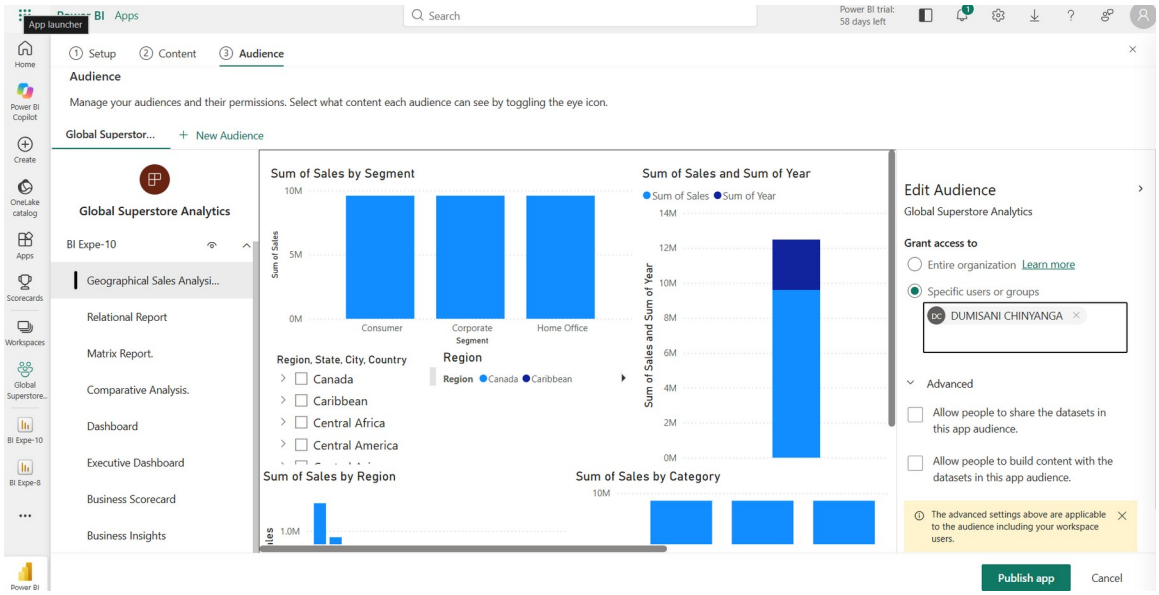


Figure 3: Creating the Global Superstore Analytics app: BI Expe-10 content with access granted to specific users.

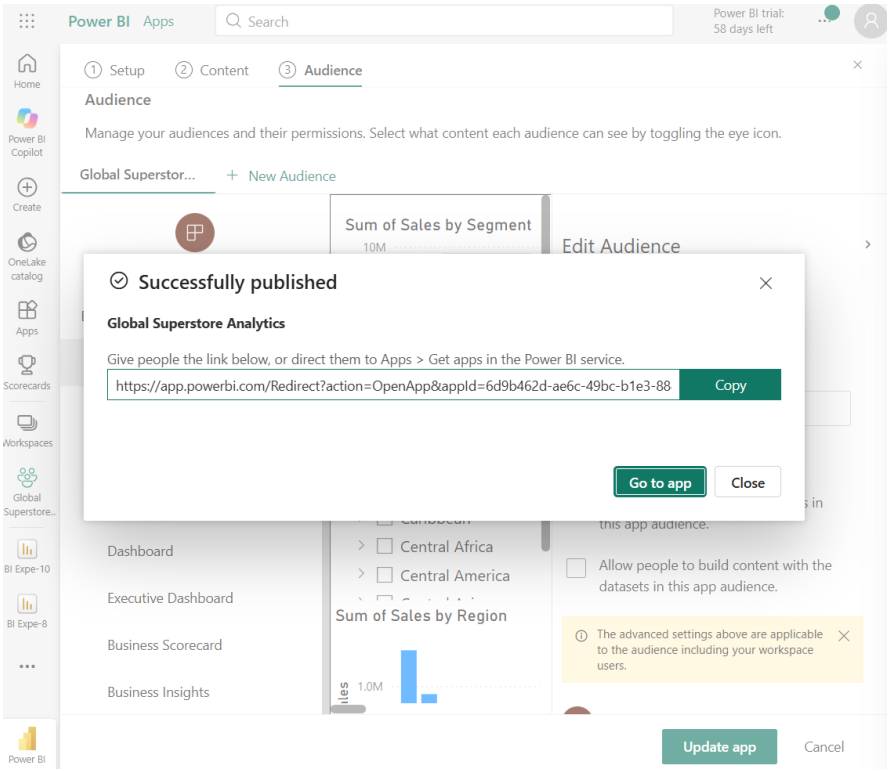


Figure 4: The Global Superstore Analytics app successfully published, with a shareable app link.

Step 7: Identified workspace roles.

Role	View	Edit / Publish	Share / App	Manage access
Admin	✓	✓	✓	✓

Role	View	Edit / Publish	Share / App	Manage access
Member	✓	✓	✓	Add lower roles
Contributor	✓	✓	✗	✗
Viewer	✓	✗	✗	✗

Step 8: Assigned responsibilities to each role.

Role	Assigned to	Responsibilities
Admin	BI Administrator (SAM FAWAZ HADI ALI AL-BASARA)	Owns both workspaces, manages access, publishes the app, runs backups and recovery
Member	BI Lead (PROD only)	Approves releases, updates the app, shares content with new audiences
Contributor	BI Developer (DUMISANI CHINYANGA in DEV)	Builds and edits reports and measures in DEV; cannot share or change access
Viewer	Tester (KONGHO SEYDOU CISSE in DEV); business users in PROD	Tests content in DEV; consumes reports in PROD with RLS applied

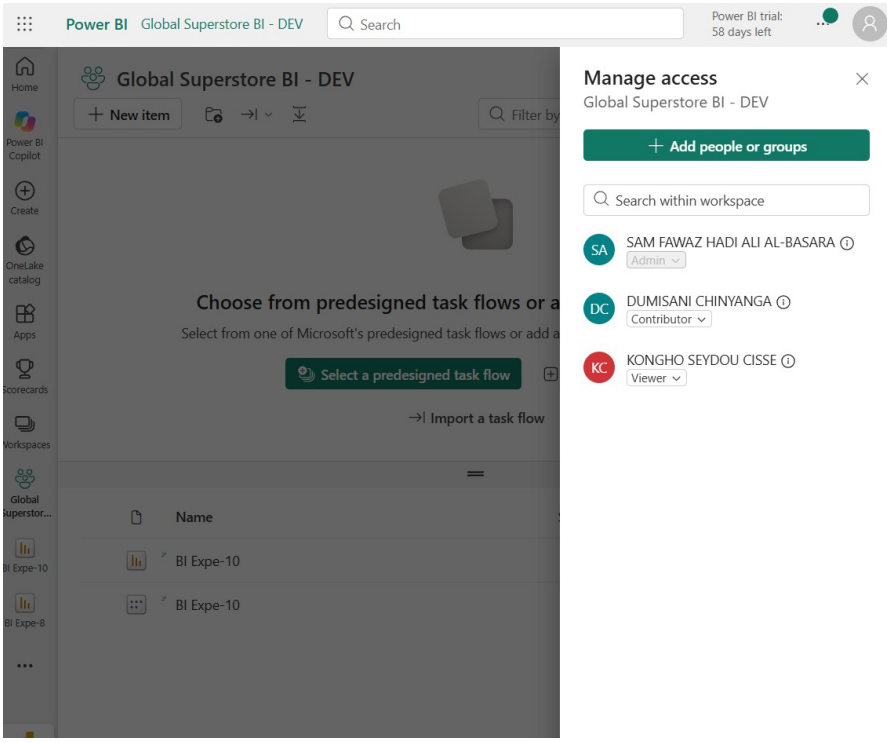


Figure 5: Manage access in Global Superstore BI - DEV: Admin, Contributor and Viewer roles assigned.

Step 9: Documented workspace governance guidelines.

- **Naming:** workspaces follow the pattern "Global Superstore BI - <stage>"; files follow BI_Experiment_<n>_<date>.

- **Separation of environments:** development happens only in DEV; PROD holds approved content only.
- **Release control:** only the BI Administrator or BI Lead publishes to PROD, after testing in DEV.
- **Least privilege:** business users never receive Contributor or higher in PROD, so RLS always applies.
- **App distribution:** end users access reports through the Global Superstore Analytics app, not the workspace.
- **Group-based access:** access is granted to security groups where possible, and reviewed every quarter.
- **Documentation:** every release records the date, version, changes and the person who approved it.

Part C: Monitoring Strategy Design

Step 10: Identified monitoring requirements and tested them in Power BI.

(a) Report performance

Performance analyzer in Power BI Desktop (View, Performance analyzer, Start recording, Refresh visuals) was run on the Executive Dashboard twice.

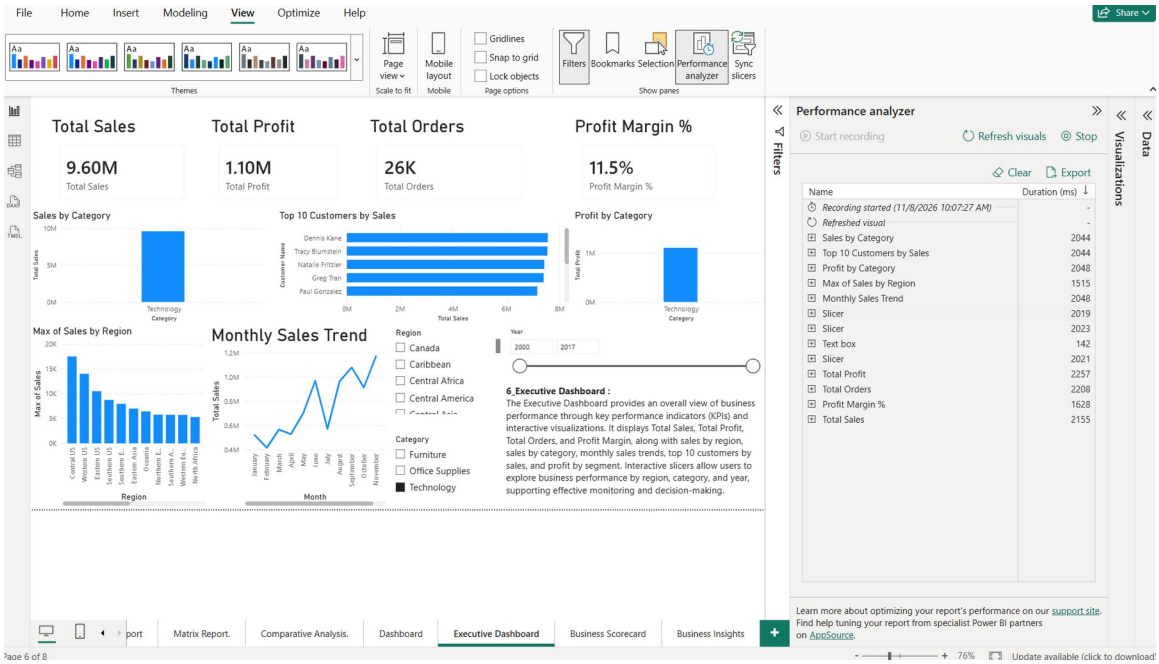


Figure 6: Performance analyzer, first refresh: load time of each visual on the Executive Dashboard.

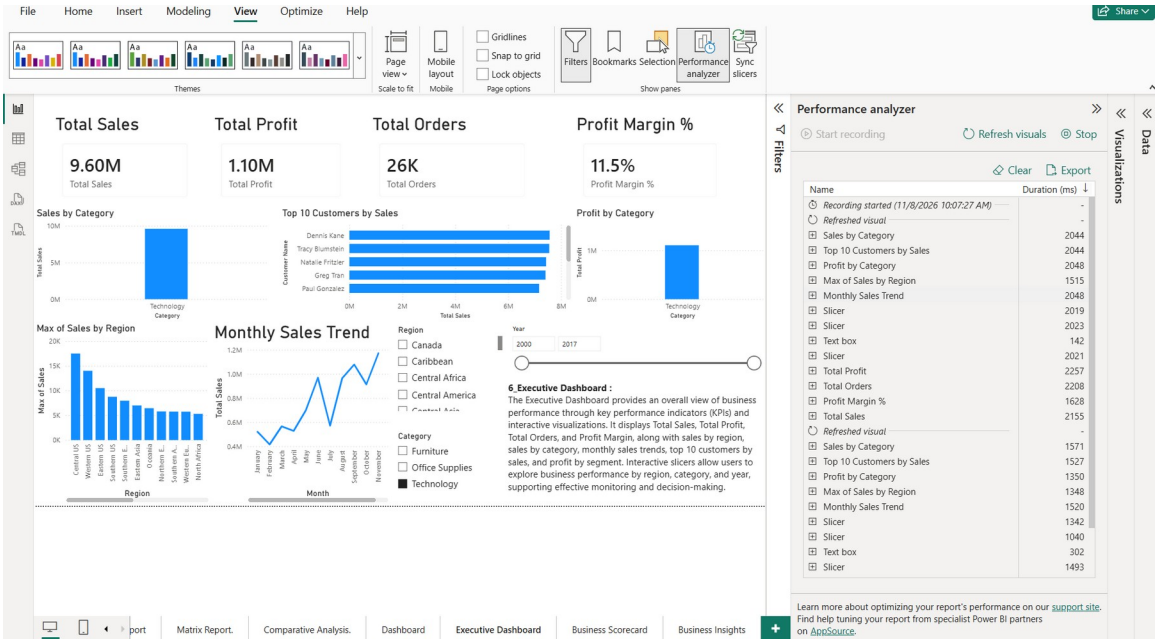


Figure 7: Performance analyzer, second refresh: visuals load faster once queries are cached.

Visual	First refresh (ms)	Second refresh (ms)
Sales by Category	2,044	1,571
Top 10 Customers by Sales	2,044	1,527
Profit by Category	2,048	1,350
Max of Sales by Region	1,515	1,348
Monthly Sales Trend	2,048	1,520
Slicers	2,019 to 2,023	1,040 to 1,493
Total Profit card (slowest)	2,257	-
Text box (fastest)	142	302

All visuals loaded in under 2.3 seconds, and the second refresh was about 20 to 50% faster. The page therefore meets a 3-second performance target. The KPI cards were the slowest visuals, so they are the first candidates for optimization if data volume grows.

(b) Dataset refresh status

Refresh now was triggered on the semantic model in PROD, and the result was checked in Settings, Refresh, Refresh history.

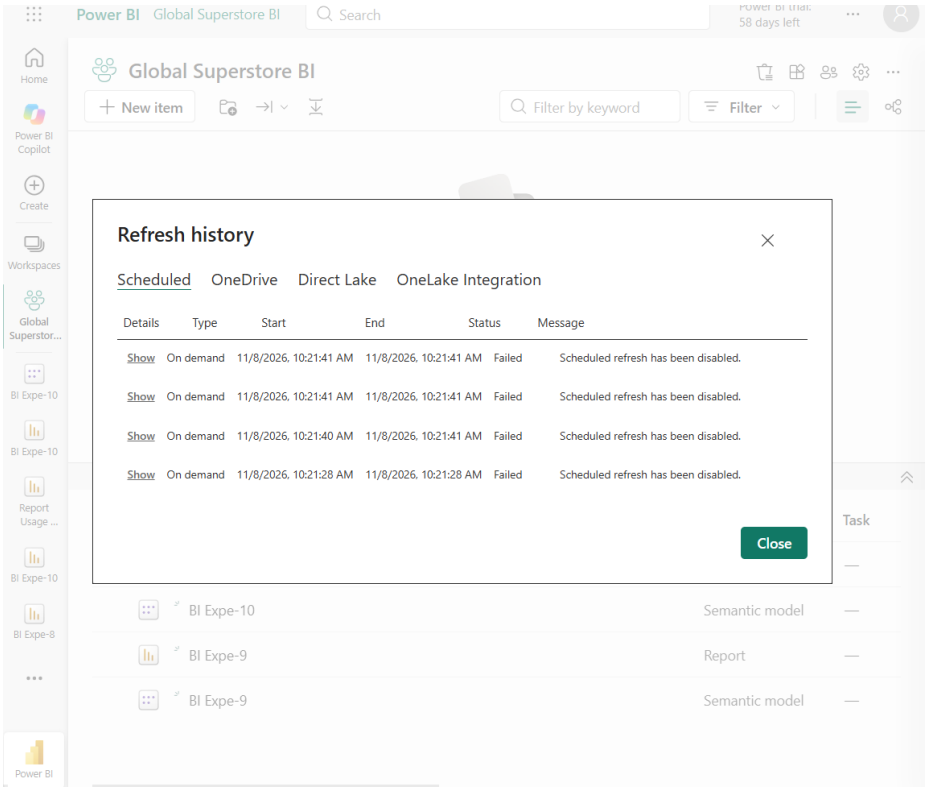


Figure 8: Refresh history: on-demand refreshes failed with the message "Scheduled refresh has been disabled".

Monitoring detected four failed on-demand refreshes on 11/8/2026 with the message "Scheduled refresh has been disabled". The source is an Excel file on a local computer, so the cloud service depends on the gateway and stored credentials to reach it. This is exactly the kind of problem a refresh monitor must catch: the report would keep showing old data while users assume it is current. Corrective action: check the gateway and data source credentials in the semantic model settings, re-enable scheduled refresh, or move the source file to OneDrive or SharePoint so it can refresh without a personal gateway.

(c) Usage statistics

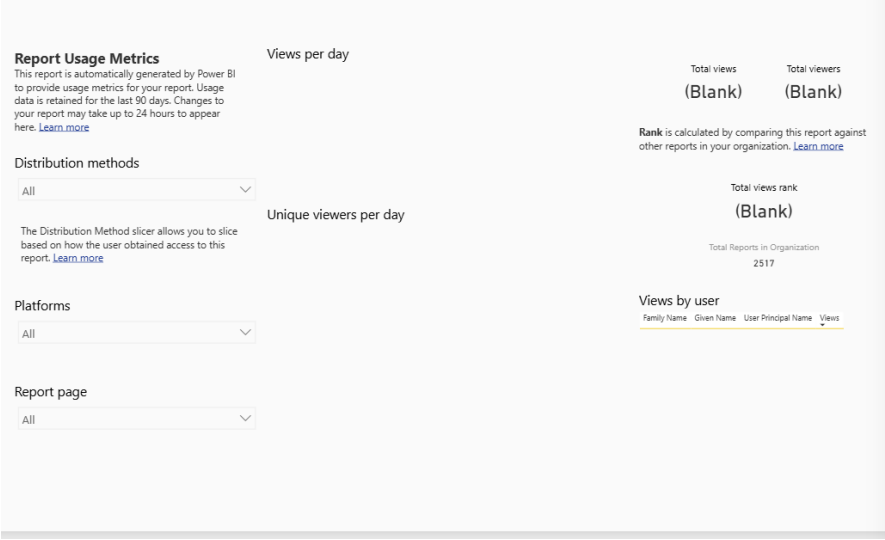


Figure 9: Report Usage Metrics: views per day, unique viewers, distribution method, platform and page filters.

The Report Usage Metrics report was generated for BI Expe-10. The figures were still blank because Power BI notes that new activity can take up to 24 hours to appear. The report shows views per day, unique viewers, views by user and a ranking against the 2,517 reports in the organization, so it will track adoption once data arrives. The lineage view already recorded early activity: 1 viewer and 3 views.

(d) User activity and dependencies

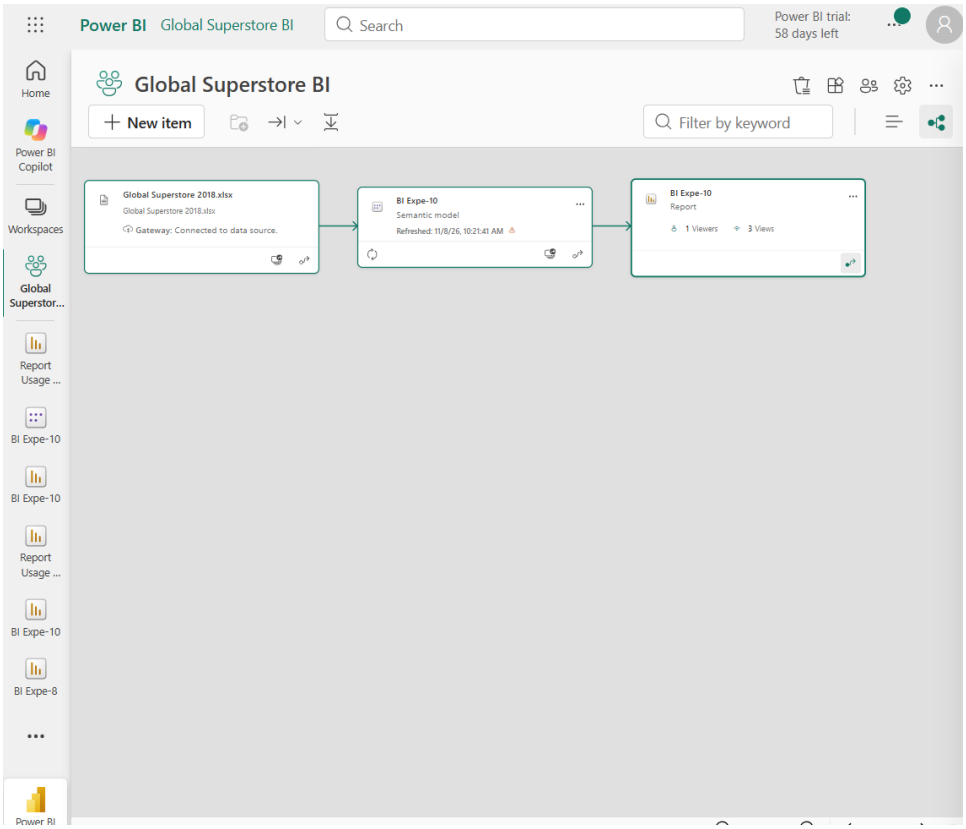


Figure 10: Lineage view of the PROD workspace: Global Superstore 2018.xlsx → BI Expe-10 semantic model → BI Expe-10 report (1 viewer, 3 views).

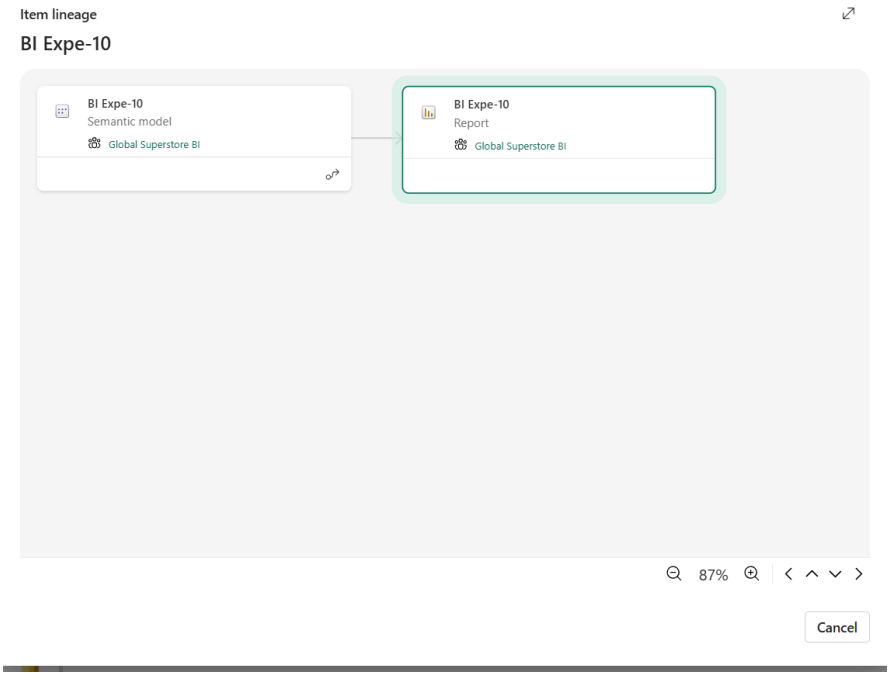


Figure 11: Item lineage for the BI Expe-10 report.

The lineage view shows that the data source is connected through the gateway, that the semantic model was last refreshed on 11/8/26 at 10:21:41 AM with a warning icon (matching the failed refresh), and that the report has 1 viewer and 3 views. It also shows the impact of a failure: if the Excel source or semantic model fails, the report and app above it are affected.

Step 11: Created the Monitoring Matrix.

Metric	Tool	Frequency	Threshold / alert condition	Responsible
Dataset refresh status	Refresh history, refresh failure email	Daily	Any failed refresh	BI Administrator
Data freshness	Lineage view (last refreshed time)	Daily	Data older than 1 day	BI Administrator
Report performance	Performance analyzer	Each release and monthly	Any visual over 3 seconds	BI Developer
Dashboard usage	Report Usage Metrics	Weekly	Sharp drop in views or unused pages	BI Lead
User activity	Usage metrics, Power BI activity log	Weekly	Unexpected users or unusual exports	BI Administrator
Access and roles	Manage access, RLS Security page	Quarterly	Users with excessive roles	BI Administrator
Data alerts and subscriptions	Subscriptions and alerts list (Exp 8)	Weekly	Failed or missing emails	BI Administrator
Gateway health	Manage connections and gateways	Daily	Gateway offline	IT Support

Part D: Backup and Restore Planning

Step 12: Identified BI components that require backup.

Component	Backup method	Contains
Report and data (.pbix)	Save As with date in Power BI Desktop; Download this file from the Service	Pages, model, measures, RLS roles and data
Model template (.pbit)	File, Export, Power BI template	Model, measures, RLS roles, pages (no data)
Data source	Copy of Global Superstore 2018.xlsx in OneDrive	Raw data
Workspace configuration	Documented access lists, RLS members and app audiences	Who has access to what
Subscriptions and alerts	Documented schedule and recipients (Exp 8)	Automated reporting setup
Project documents	Reports, presentation and matrices in OneDrive	Documentation

Step 13: Developed the Backup Schedule.

Backup	Frequency	What is saved	Location	Retention
Daily Dataset Backup	Daily, end of day	BI_Experiment_10_<date>.pbix with data	BI_Backups / Daily	Last 7 copies
Weekly Report Backup	Every Friday	.pbit template of model, measures, roles and pages	BI_Backups / Weekly	Last 4 copies
Monthly Workspace Archive	Last working day of month	Downloaded .pbix from PROD, source Excel file, access and RLS documentation	BI_Backups / Monthly	12 months

The backup folders were created in OneDrive, so every backup is stored in the cloud as well as locally and keeps OneDrive version history.

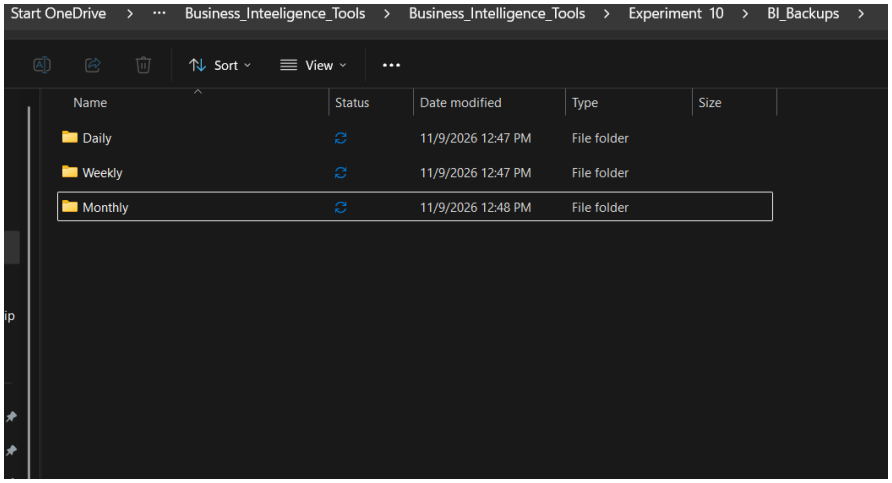


Figure 12: BI_Backups folder in OneDrive with Daily, Weekly and Monthly backup folders.

Step 14: Designed the Restore Procedure.

Case 1: Accidental deletion of a report or dashboard

1. Confirm what was deleted and when (report, semantic model, or both).
2. Open the workspace, click Recycle bin, select the deleted item and click Restore.
3. If the item is not in the Recycle bin, open the latest Daily .pbix backup in Power BI Desktop and Publish it to the same workspace.
4. Re-check RLS role members on the semantic model and test with Test as role.
5. Update the app so the restored report is included, and recreate any subscriptions or alerts.
6. Inform users that the content is available again.

Case 2: Report corruption or a bad release

1. Stop further changes in PROD.
2. Open the last known good .pbix from Daily, or the .pbit from Weekly and refresh the data.
3. Publish to DEV and test, then publish to PROD with Replace.

4. Record the incident and the cause in the release log.

Case 3: System or data source failure

1. Check the refresh history and gateway status to find the failing component.
2. Restore the source Excel file from the OneDrive copy or version history if it is missing or damaged.
3. Fix gateway or credential settings and run Refresh now.
4. If the workspace itself is lost, create a new workspace, republish from the Monthly archive, and reapply access, RLS members and the app from the documentation.

Part E: Disaster Recovery Scenario

Step 15: Analyzed the scenario: "A critical executive dashboard is accidentally deleted one day before a management meeting."

Aspect	Analysis
Asset affected	Executive Dashboard report used by the CEO and management
Business impact	High: management cannot review KPIs, targets and insights before the meeting
Linked items affected	App content, subscriptions and alerts linked to the report
Time available	Less than 1 day, so recovery must be completed within the 1 hour RTO
Data loss risk	Low: daily backups limit any loss to at most 1 day (RPO)

The scenario was tested safely with a copy of the report named **Executive Dashboard DR Test**, so production content was not affected. The copy was published, deleted to simulate the accident, and then restored.

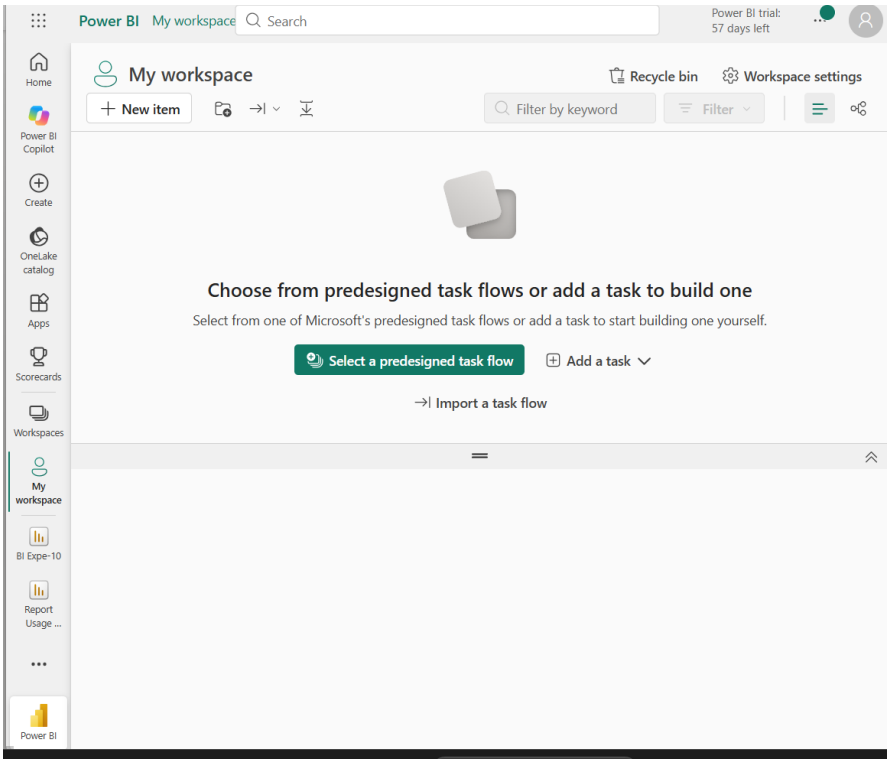


Figure 13: Disaster simulation: the workspace after the Executive Dashboard DR Test report was deleted.

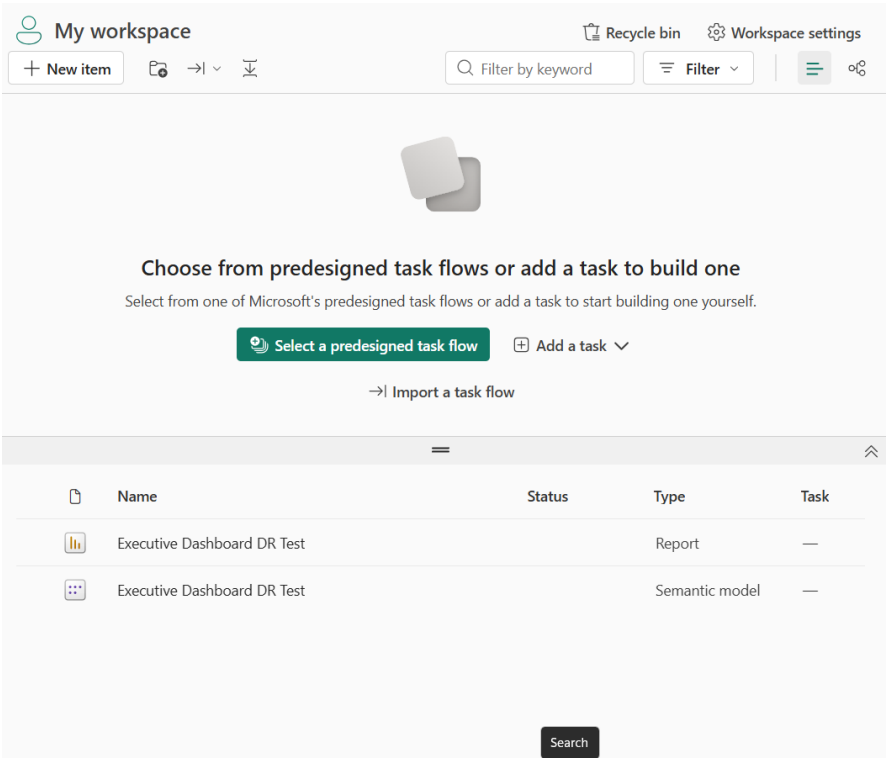


Figure 14: Recovery: the Executive Dashboard DR Test report and semantic model available again after the restore.

Step 16: Explained how backup and restore recover the solution.

Time	Action	Backup used
0 to 10 min	User or monitoring reports the missing dashboard; BI Administrator confirms the deletion	-
10 to 20 min	Restore the report from the workspace Recycle bin	Recycle bin
20 to 40 min	If not in the Recycle bin: publish the latest Daily .pbix to PROD	Daily .pbix
40 to 50 min	Verify all pages, KPIs (9.60M sales, 1.10M profit) and RLS with Test as role	-
50 to 60 min	Update the app, recreate subscriptions and alerts, notify management	Documentation

Because the full report, model, measures and RLS roles are inside the backup .pbix, the dashboard can be rebuilt exactly as it was without redesigning anything, well before the meeting.

Step 17: Documented actions required for business continuity.

- **Before the incident:** keep daily, weekly and monthly backups in OneDrive; document access lists, RLS members, app audiences and subscriptions.
- **Prevent deletion:** keep business users as Viewers, so only the BI team can delete content.
- **Fallback for the meeting:** export the Executive Dashboard to PDF or PowerPoint before every management meeting, so a static copy is always available.
- **During recovery:** communicate the expected recovery time to management and follow the restore procedure in order.
- **After recovery:** test the restored report, record the incident, find the cause, and update the procedure if needed.
- **Regular testing:** run a disaster recovery drill every quarter, as done in this experiment with the DR Test report.

Part F: Summary of Deliverables

Deliverable	Where in this report
BI Deployment Plan	Part A, Steps 1 to 5
Workspace and Role Structure Diagram	Part B, Step 6 (Figure 1) and Steps 7 to 9
Monitoring Matrix	Part C, Step 11
Backup and Recovery Plan	Part D, Steps 12 to 14
Disaster Recovery Report	Part E, Steps 15 to 17

7. RESULT

A complete BI deployment, monitoring, backup, and recovery strategy was successfully designed and demonstrated for the Global Superstore BI solution using Microsoft Power BI. BI assets from

Experiments 5 to 9 were identified and a phased deployment plan was prepared. A DEV and PROD workspace structure was created with Admin, Contributor and Viewer roles, and the solution was deployed to users through the Global Superstore Analytics app.

Monitoring was carried out with Performance analyzer (all visuals under 2.3 seconds), refresh history (which detected failed refreshes), usage metrics and lineage view, and a Monitoring Matrix was defined. Daily, weekly and monthly backups were planned in OneDrive, restore procedures were written, and the accidental-deletion scenario was tested by deleting and restoring a DR test report, ensuring reliable and secure BI operations.

8. LEARNING OUTCOMES

- Analyze deployment requirements for Business Intelligence solutions. (CO4-BT4)
- Design workspace structures and governance models for BI systems. (CO5-BT6)
- Evaluate monitoring requirements for deployed BI solutions. (CO4-BT4)
- Develop backup and recovery strategies for business continuity. (CO5-BT6)
- Design complete deployment and maintenance plans for organizational BI environments. (CO5-BT6)